Peter Mao

Department of Mathematics, Drexel University
Korman Center, Room 221, 15 S. 33rd Street, Philadelphia, PA 19104
pm929@drexel.edu · maopeter.github.io

Education

Drexel University 2022 – present
Ph.D. in Mathematics (expected June 2027)
Advisor: Darij Grinberg
M.S. in Mathematics, 2024

Cornell University 2018 – 2022
B.A. in Mathematics, *cum laude*, 2022

Research Interests

Algebraic combinatorics, dynamical algebraic combinatorics,
symmetric group algebras, permutation statistics.

Publications and Preprints

1. *Foatic homomesy.* Preprint (2026). arXiv:2610.40882. Latest version.
2. (with Darij Grinberg) *Necklaces over a group with identity product. Journal of Integer Sequences* **29** (2026), Article 26.4.6. Journal version. arXiv:2405.08937.

Talks

- *Foatic homomesy: telescoping proofs for orbit averages on S_n.* October 2026
  CAGE Seminar (Combinatorics, Algebra, and Geometry), Drexel University.
  Slides.
- *Foatic homomesy: telescoping proofs for orbit averages on S_n.* May 2026
  SIAM Student Chapter, Drexel University
- *Necklaces over a group with identity product: counting and bijective connections.* November 2024
  SIAM Student Chapter, Drexel University

Conferences and Workshops

- Mid-Atlantic Algebra, Geometry, and Combinatorics Workshop (MAAGC), October 2026
  Virginia Commonwealth University
- Formal Power Series and Algebraic Combinatorics (FPSAC 2026), July 2026
  University of Washington
- Triangle Lectures in Combinatorics, Wake Forest University April 2026

Teaching

Instructor of record, Drexel University

- MATH 122, Calculus II — Fall 2026
- MATH 221, Discrete Mathematics (mastery-based testing redesign) — Summer 2026
- MATH 101, Business Pre-Calculus (*Introduction to Analysis I*) — Fall 2024
- MATH 221, Discrete Mathematics — Summer 2024

Teaching assistant, Drexel University

- MATH 311, Probability and Statistics I — Winter 2024, Spring 2025, Spring 2026
- MATH 200, Multivariate Calculus — Spring 2024, Winter 2025, Fall 2025
- MATH 121, Calculus I — Winter 2026
- MATH 102, Business Calculus I (*Introduction to Analysis II*) — Fall 2022 – Spring 2023

Tutor, Math Resource Center, Drexel University — Fall 2022 – present

Undergraduate teaching assistant, introductory physics labs, Cornell University — Fall 2019

References

Available upon request.